

Concept-Oriented Numerical Practice

Gravitation and Electrostatics

Name:

Suggested Time: 90 minutes

Instructions:

- Show all relevant steps and reasoning.
 - Use SI units throughout.
 - Numerical answers without explanation may receive reduced credit.
 - Read the hint only if you are stuck.
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Problem 1

Two particles of masses $m_1 = 3.0\text{ kg}$ and $m_2 = 12.0\text{ kg}$ are fixed on the y -axis at positions $y = 0$ and $y = 6.0\text{ m}$, respectively.

A third particle of mass m_3 is placed somewhere on the y -axis.

Question: Find the position where the net gravitational force on m_3 is zero.

Hint: Determine where the forces can oppose each other, then equate their magnitudes.

Problem 2

Assume Earth has uniform density.

An object of mass 2.0 kg experiences a gravitational force of 700 N at Earth's surface.

Question: What gravitational force does the object experience when it is located at a distance $0.40R$ from Earth's center?

Hint: Inside a uniform sphere, gravitational force is proportional to the distance from the center.

Problem 3

Three charges are arranged as follows:

- $q_1 = +5.0\text{ }\mu\text{C}$ at the origin,
- $q_2 = -5.0\text{ }\mu\text{C}$ at $(0.20\text{ m}, 0)$,
- $q_3 = +5.0\text{ }\mu\text{C}$ at $(0.20\text{ m}, 0.30\text{ m})$.

Question: Find the magnitude and direction of the net force acting on q_1 .

Hint: Resolve each force into components before adding.

Problem 4

Four identical charges $+2.0\text{ }\mu\text{C}$ are placed at the corners of a square of side length 0.40 m . A fifth identical charge is placed at the center of the square.

Question: Calculate the magnitude of the net electric force on the central charge.

Hint: Use symmetry to determine how forces from opposite corners combine.

Problem 5

A point charge produces an electric field at a location in space.

A charge $+2.0\text{ }\mu\text{C}$ placed at that location experiences a force of 0.30 N .

Question:

- Find the magnitude of the electric field at that location.
- What force would a charge $-1.0\text{ }\mu\text{C}$ experience if placed there?

Hint: The electric field depends only on the source charge, not the test charge.

Problem 6

An electric dipole consists of charges $\pm 4.0\text{ }\mu\text{C}$ separated by 6.0 mm .

The dipole is placed in a uniform electric field of magnitude $E = 2.0 \times 10^5\text{ N/C}$ and makes an angle of 30° with the field.

Question: Find the magnitude of the torque acting on the dipole.

Hint: Use the torque formula involving the dipole moment and $\sin\theta$.

Problem 7

A very long straight rod carries a uniform linear charge density $\lambda = 4.0 \times 10^{-6} \text{ C/m}$.

Question: Find the magnitude of the electric field at a distance 0.20 m from the rod.

Hint: Model the rod as infinitely long and choose a cylindrical Gaussian surface.

Problem 8

A non-conducting spherical shell has inner radius $a = 2.0 \text{ cm}$ and outer radius $b = 5.0 \text{ cm}$. The shell has uniform volume charge density $\rho = 1.5 \times 10^{-2} \text{ C/m}^3$.

A point charge $q = -6.0 \text{ }\mu\text{C}$ is at the center.

Question: Find the magnitude of the electric field at $r = 4.0 \text{ cm}$.

Hint: Only charge enclosed within the Gaussian surface contributes.

Problem 9

A conducting spherical shell has a total charge of $+10 \text{ }\mu\text{C}$.

A point charge $+3.0 \text{ }\mu\text{C}$ is placed at the center of a cavity inside the conductor.

Question:

- (a) What charge is induced on the inner surface?
- (b) What charge appears on the outer surface?

Hint: The electric field inside the conducting material must be zero.

Problem 10

A spherical Gaussian surface of radius 0.50 m encloses a net charge of $-8.0 \text{ }\mu\text{C}$.

Question:

- (a) Calculate the electric flux through the surface.
- (b) Does the shape of the surface affect the flux? Explain briefly.

Hint: Electric flux depends only on the total enclosed charge.